

Madhusudan Bipinbhai Mandaliya

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GIS Specialist with expertise in geographic information systems, spatial data analysis, and mapping. I am proficient in GIS software, data visualization, and geospatial technologies to deliver impactful solutions. Skilled in managing complex projects, ensuring data accuracy, and driving sustainable development. Known for effective communication, teamwork, and adaptability in achieving project goals. Passionate about leveraging GIS to improve community quality of life and support data-driven decision-making.

EDUCATION

Master of Science, Geographic Information Science and Cartography

The University of Texas at Dallas, Richardson, TX

GPA 3.93/4.0

January 2023 – December 2024

Master's, Urban and Regional Planning

The Maharaja Sayajirao University of Baroda, Gujarat, India

GPA 3.91/4.0

August 2017 – April 2019

Bachelor of Technology, Civil Engineering

Gujarat Technological University, Gujarat, India

CGPA 8.07/10.0

July 2013 – June 2017

PROFESSIONAL EXPERIENCE

Geographic Information Systems (GIS) Co-op | FedEx Ground | Moon Township, PA

August 2024 – December 2024

- Developed district-specific Helper Maps with advanced GIS layers and filters, enabling teams to identify high-priority zones and optimize route planning, achieving an average 5% cost reduction.
- Designed and implemented geofence boundaries for facilities using ArcGIS Pro, leveraging georeferenced CAD layers and satellite imagery to enhance package tracking accuracy and streamline facility operations.
- Conducted GPS breadcrumb data analysis to address missing street segments, improving routing efficiency by enhancing and validating street network datasets through spatial editing and integration with HERE Maps.
- Developed interactive web applications to visualize HERE road closure segments within ZIP codes.
- Conducted RAAS-based route optimization, validating delivery parameters and enhancing delivery station modeling.

Urban Planner | Planning Solutions | Vadodara, Gujarat

May 2019 – April 2022

- Developed GIS-based Master Plans in collaboration with the Indian government under the AMRUT scheme, using ArcGIS to create georeferenced base maps and land use maps, ensuring data accuracy for urban planning.
- Constructed a 1:4000 scale urban database by integrating multiple data sources, adhering to design standards to support comprehensive project planning and decision-making.
- Prepared GIS figures for reporting, planning documents, preparing digital maps for desktop review and public access.
- Managed Geodatabase while gathering, compiling, migrating, and converting datasets from various internal and external sources, including CADD to GIS conversions.
- Utilized GIS software and predictive modeling technologies for urban planning and infrastructure development.

Jr. Town Planner (Intern) | Surat Urban Development Authority | Surat, Gujarat

May 2018 – June 2018

- Collaborated with the Town Planning Officer on the Surat Development Plan – 2035 and 8 Town Planning Schemes.
- Assisted in developing the 90-meter Outer Ring Road, site inspections, and geospatial data collection.

SKILLS AND CERTIFICATIONS

Technical Skills: ArcGIS Desktop, ArcGIS Pro, Experience Builder, Story Map, Dashboard, Survey123, Field Survey Map, Model Builder, Adobe Photoshop, Illustrator, InDesign, AutoCAD, Revit, Microsoft 365, QGIS

Programming: Python, JavaScript for ArcGIS API, HTML

Certifications: Fundamentals of GIS, AutoCAD 2D and 3D, ArcGIS: From Beginner to Pro, Revit Architecture, Spatial Analysis and Geospatial Data Science in Python, GIS Data Formats, Design and Quality

PROJECTS

Safety Map: Visualizing Safe Routes with GIS using Python

May 2024

- Developed a GIS-based system using Python and the Network Analyst geoprocessing tool to calculate safety scores for route segments based on accident data, prioritizing safer routes by integrating 3 stops into the analysis.
- Designed a map to visualize high-risk areas and help users identify safe travel routes with customizable preferences.

Effect of Land Use and Perceived Traffic Accident Risk on Selection of Public Transit as a Travel Mode – Vadodara

April 2019

- Applied GIS data standards, editing, and updating database to ensure data integrity. Created bus routes and bus stops feature layers using ArcGIS Field Maps, mapping over 150 bus stops and 32 routes for further analysis.
- Developed land use maps near high and low transit ridership areas to analyze their impact.